INVERTIS UNIVERSITY, BAREILLY

B.Tech CSE – Third Year | DBMS & SQL Lab (BCS552)

EXPERIMENT NO. 3

1. Experiment Title:

Selecting Columns and Rows with WHERE

2. Aim:

Display selected columns and retrieve only records that satisfy specified conditions.

3. Software / Tools Required:

SQL database system (such as MySQL, PostgreSQL, or SQLite where supported), SQL editor/IDE, and a computer.

4. Theory:

SELECT chooses the columns to display. WHERE filters rows according to a condition. Comparison operators such as > and = are used to test values.

5. Algorithm / Procedure:

1. Create the student table and insert the four given records.
2. Select only name and marks.
3. Select all columns for students whose marks are greater than 60.
4. Select the name of the student whose roll number is 103.

6. Program / Queries / Commands:

```
CREATE TABLE student (
    roll_no INT,
    name VARCHAR(30),
    marks INT
);

INSERT INTO student (roll_no, name, marks)
VALUES
    (101, 'Asha', 78),
    (102, 'Rohan', 64),
    (103, 'Neha', 88),
    (104, 'Deepak', 52);

-- Task 1
SELECT name, marks FROM student;

-- Task 2
SELECT * FROM student
WHERE marks > 60;

-- Task 3
SELECT name FROM student
WHERE roll_no = 103;
```

7. Input:

(101, 'Asha', 78), (102, 'Rohan', 64), (103, 'Neha', 88), (104, 'Deepak', 52)

8. Output:

Expected output (column spacing and display format may differ in your SQL tool):

Task 1: name	marks	

Asha	78	
Rohan	64	
Neha	88	
Deepak	52	
Task 2: roll_no	name	marks
101	Asha	78
102	Rohan	64
103	Neha	88
Task 3: name		
Neha		

9. Result:

The SQL statements for Experiment 3 were prepared to perform the stated tasks. On executing the setup and queries successfully, the expected results shown above should be obtained.

10. Viva-Voce Questions:

1. What is the purpose of SELECT?
2. What does WHERE do?
3. What is the difference between SELECT * and SELECT name?
4. Which operator checks whether marks are greater than 60?